Species-Aware Review: Bacterial ISC Iron–Sulfur Cluster Assembly in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Commissioned topic:** `bacterial_isc_iron_sulfur_cluster_assembly` **Resolved bucket:** KEGG `ppu04122` "Sulfur relay system" (22 candidate genes) **Prepared:** Iteration 3 of a 3-iteration autonomous review

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1. Executive summary

The review topic and the resolved gene bucket describe two different modules. The commissioned topic is *ISC iron–sulfur cluster assembly* (the `iscRSUA–hscBA–fdx` machinery that builds [2Fe-2S]/[4Fe-4S] cofactors), but the resolved KEGG bucket `ppu04122` is the **"Sulfur relay system"** — a distinct KEGG map covering **tRNA thiolation** (2-thiouridine, 4-thiouridine) and **molybdenum-cofactor (Moco)/thiamine sulfur transfer**. The only node shared by both is the cysteine desulfurase **IscS (PP_0842)**, which is the sulfur-mobilizing hub feeding both processes. This is the single most important curation conclusion: **the bucket is the wrong gene source for an ISC review** and needs to be split.

Key results (all from UniProt proteome UP000000556 + InterPro/Pfam signatures, corroborated by literature):

1. **The ISC module IS satisfiable in PSEPK.** A complete, contiguous canonical operon exists at **PP_0841–PP_0847** = `iscR–iscS–iscU–iscA–hscB–hscA–fdx`. Six of these seven genes are **absent from the candidate list** (only `iscS` PP_0842 is included).
2. **No SUF system.** `sufABCDSE` is absent; ISC is the **sole** Fe–S biogenesis machinery in PSEPK (contrast with *E. coli*, which has both ISC and stress-inducible SUF). A SUF module step should be marked `not_expected_in_target_taxon`.
3. **The `ppu04122` bucket itself is largely well-populated** for the sulfur-relay/Moco processes it actually represents (tRNA 2-thiouridine relay complete; Moco and thiamine sulfur

carriers present), **but contains over-propagated MoaA annotations** (PP_1969, PP_2482) and desulfurase/TusA paralog ambiguity.

Bottom line for curators: mark ISC as `covered` using PP_0841–PP_0847 (add the 6 missing genes), mark SUF `not_expected_in_target_taxon`, and flag `ppu04122` as `module_needs_revision` (split ISC vs. sulfur-relay vs. Moco).

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2. Target-organism pathway definition

ISC iron–sulfur cluster assembly (the commissioned topic). The housekeeping bacterial pathway that assembles nascent [2Fe-2S]/[4Fe-4S] clusters and delivers them to apoproteins. Mechanistically: iron binds the scaffold **IscU**; the cysteine desulfurase **IscS** inserts persulfide sulfur; the ferredoxin **Fdx** reduces persulfide to sulfide, producing a [1Fe-1S] precursor; two precursors fuse on an IscU dimer to a bridging [2Fe-2S] cluster; the **HscA/HscB** chaperone/co-chaperone pair drives ATP-dependent cluster transfer; **IscA** acts as an A-type carrier/alternate scaffold; **IscR** (Rrf2-family, [2Fe-2S]-sensing) transcriptionally regulates the operon (P 39870763; P 39632806).

What to keep separate (neighbouring maps / broad overviews): - **KEGG `ppu04122` "Sulfur relay system"** — tRNA thiolation + Moco/thiamine sulfur transfer. *This is the resolved bucket, and it is NOT the ISC pathway.* Keep separate. - **KEGG `ppu00730` "Thiamine metabolism"** and Moco biosynthesis (KEGG module M00880) — downstream sulfur users. - **SUF system** (`sufABCDSE`) — the alternative Fe–S assembly pathway; **absent** in PSEPK. - **CIA / NIF systems** — eukaryotic cytosolic and nitrogen-fixation-specific Fe–S systems; not applicable.

Alternate names / DB definitions: ISC = "iron–sulfur cluster" system; operon genes also appear as `isc`/`nifU`-like (IscU carries the NifU_N domain) and `hsc` (Hsp70-class chaperones). KEGG places `iscS` under sulfur/relay metabolism, which is why it — and not the rest of the operon — leaked into the `ppu04122` bucket.

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3. Expected step model (ISC) and status in PSEPK

ISC step	Expected gene	PSEPK locus	UniProt	Evidence (domain signature)	Status
Cysteine desulfurase (S donor)	iscS	**PP_0842**	Q88PK8	Aminotran_5; Cys desulfurase; EC 2.8.1.7	`covered` (in bucket)
Scaffold	iscU	**PP_0843**	Q88PK7	NifU_N; IPR011339 ISCU	`covered` (missing from bucket)
A-type carrier	iscA	**PP_0844**	Q88PK6	Fe-S_biosyn; IPR011302 IscA_proteobact	`covered` (missing)
Co-chaperone (J-protein)	hscB	**PP_0845**	Q88PK5	DnaJ + HSCB_C; IPR004640 HscB	`covered` (missing)
Chaperone (Hsp70)	hscA	**PP_0846**	Q88PK4	HSP70; IPR042039 HscA_NBD	`covered` (missing)
Ferredoxin (e^- donor)	fdx	**PP_0847**	Q88PK3	Fer2; IPR011536 **Fdx_isc**	`covered` (missing)
Regulator ([2Fe-2S] sensor)	iscR	**PP_0841**	Q88PK9	Rrf2; IPR010242 **TF_HTH_IscR**	`covered` (missing)
Alternative assembly system	suf operon	—	—	0 hits for sufABCDSE	`not_expected_in_target_taxon`

Gene order PP_0841→PP_0847 reproduces the textbook `iscR-iscS-iscU-iscA-hscB-hscA-fdx` operon, giving strong **direct genomic (species-specific)** support that the ISC module is complete and satisfiable.

4. Candidate genes and evidence (the `ppu04122` bucket, 22 genes)

Evidence type is **homology/domain-signature** unless noted; all loci are direct PSEPK sequences from proteome UP000000556. None of the candidate genes below are ISC-assembly genes except `iscS`.

A. Sulfur donor / hub - iscS PP_0842 (Q88PK8) — Cysteine desulfurase, EC 2.8.1.7. The shared hub for ISC assembly, tRNA thiolation, and Moco/thiamine. High confidence. *Only bucket gene that is also an ISC gene.*

B. tRNA 2-thiouridine (mnm^5s^2U34) relay — complete - tusA PP_2116 (Q88L21) — TusA sulfur carrier (IPR022931, specific). High confidence. - **tusB PP_3995 / tusC PP_3994 / tusD PP_3993 (Q88FT7/FT8/FT9)** — TusBCD (DsrEFH-like: DsrH/DsrF/DsrE). Clustered PP_3993–3996 → strong operon support. Note the DsrEFH family also functions in sulfur oxidation in other taxa, but in this non-sulfur-oxidizer the genomic context supports the TusBCD (tRNA) assignment. - **tusE PP_3996 (Q88FT6)** — DsrC/TusE (IPR007453). High confidence. - **mnmA PP_4014 (Q88FR9)** — tRNA 2-thiouridylase, EC 2.8.1.13. Terminal transferase. High confidence. Relay architecture (IscS→TusA→TusBCD→TusE→MnmA) is directly supported by *E. coli* biochemistry
(P 16387657).

C. 4-thiouridine / thiamine sulfur transfer - thiI PP_5045 (Q88CY4) — tRNA sulfurtransferase / 4-thiouridine synthase, EC 2.8.1.4. High confidence. - **thiS PP_5105 (Q88CS5)** — ThiS sulfur carrier (ThiS/MoaD-like, β-grasp). High confidence for thiamine branch.

D. Molybdenum-cofactor (Moco) sulfur transfer & synthesis - moeB PP_0735 (Q88PW3) — MoaD adenylyltransferase, EC 2.7.7.80. High confidence. - **moaC PP_1292 (Q88NC0)** — cyclic pyranopterin monophosphate synthase, EC 4.6.1.17. High confidence. - **moaD PP_1293 (Q88NB9) / moaE PP_1294 (Q88NB8)** — molybdopterin synthase (sulfur carrier + catalytic, EC 2.8.1.12). High confidence; MoaD is a β-grasp sulfur carrier analogous to ThiS. - **moaA PP_4597 (Q88E69)** — GTP 3',8-cyclase, EC 4.1.99.22; carries the **MoaA-specific** IPR013483 + IPR000385. High confidence — the bona fide MoaA. - **moaB PP_2122 (Q88L15) & PP_4600 (Q88E67)** — two MoaB paralogs (IPR013484 MoaB_proteobac). Genuine paralogy.

E. Over-propagated / ambiguous within the bucket - PP_1969 (Q88LG4) & PP_2482 (Q88K11) — annotated "Molybdenum cofactor biosynthesis protein A" but carry only generic **MoaA-like** radical-SAM signatures (IPR040064, IPR010505), **no EC**, and **not** the MoaA-specific IPR013483 held by PP_4597. **Likely over-propagated** → `candidate_uncertain`. - **tusA-I PP_1233 (Q88NH6)** — only generic TusA-like (IPR001455), not the specific TusA IPR022931 of PP_2116.

Possible non-relay TusA/YeeD-like paralog → `candidate_uncertain`. - **iscS-II PP_2435 (Q88K56)** — a second generic cysteine desulfurase (IPR016454). Role unassigned. - **rhdA PP_4907 (Q88DC0)** — single-domain rhodanese (thiosulfate sulfurtransferase), broad EC 2.8.1.-. Generic. - **sseA PP_5118 (Q88CR2)** — 3-mercaptopyruvate sulfurtransferase (TST/MPST-like), EC 2.8.1.2. Generic.

Not in the bucket but present in the genome (SufS-clade): - CsdA (Q9Z408) — "Probable cysteine desulfurase," SufS/CsdA clade (**IPR010970 Cys_dSase_SufS**), present **despite absence of a SUF scaffold**; likely a Moco/tRNA CsdA-type desulfurase. Needs role assignment.

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5. Gaps, ambiguities, and likely over-annotations

1. **ISC genes missing from the bucket (biggest gap).** iscR/iscU/iscA/hscB/hscA/fdx (PP_0841, PP_0843–PP_0847) are all present in PSEPK but absent from `ppu04122`. The topic-vs-bucket mismatch, not a biological gap, is the cause.
2. **MoaA over-propagation.** PP_1969, PP_2482 lack MoaA-specific signatures/EC — probably not GTP 3',8-cyclases.
3. **Desulfurase paralogy (3 enzymes).** IscS (PP_0842), PP_2435, and SufS-clade CsdA (Q9Z408). Broad EC 2.8.1.7 invites over-propagation; specific roles (ISC vs. relay vs. Moco/CsdA) are not resolvable from annotation alone.
4. **TusA paralogy.** PP_2116 (bona fide) vs. PP_1233 (generic TusA-like).
5. **DsrEFH ambiguity.** TusBCD (PP_3993–3995) belong to the DsrEFH superfamily used for sulfur oxidation in other lineages; assignment as tRNA-thiolation TusBCD rests on genomic clustering and homology, not direct PSEPK assay.
6. **No SUF backup.** Fe–S biogenesis depends solely on ISC — a genuine lineage feature, not a metadata gap.

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6. Module and GO-curation recommendations

- **ISC Fe–S assembly module:** mark **`covered`**. Add/assert PP_0841 (iscR), PP_0843 (iscU), PP_0844 (iscA), PP_0845 (hscB), PP_0846 (hscA), PP_0847 (fdx); iscS = PP_0842. Consider creating a dedicated **ISC module document** (`iscRSUA-hscBA-fdx`) rather than relying on the KEGG sulfur-relay bucket.

- **SUF assembly step:** mark `not_expected_in_target_taxon` (no sufABCDSE).
- `ppu04122` **bucket:** mark `module_needs_revision` — it conflates three biologically distinct modules (ISC assembly; tRNA thiolation relay; Moco/thiamine sulfur transfer). Recommend splitting into separate module docs.
- **tRNA 2-thiouridine relay** (if curated as its own module): `covered` (IscS-TusA-TusBCD-TusE-MnmA).
- **Moco sulfur-transfer/synthesis** (if curated): core steps `covered` (moeB, moaC, moaD/E, moaA PP_4597, moaB).
- `candidate_uncertain`: PP_1969, PP_2482 (MoaA-like), PP_1233 (TusA-like), PP_2435 (desulfurase), CsdA Q9Z408).
- **GO requests:** likely none new for ISC (well-covered by GO:0016226 iron–sulfur cluster assembly and gene-specific terms). Focus curation effort on disambiguating the three desulfurases and the MoaA-like radical-SAM paralogs.

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7. Genes to promote to full `fetch-gene` review

High priority (add to ISC module; currently missing): 1. PP_0841 iscR (Q88PK9) — regulator 2. PP_0843 iscU (Q88PK7) — scaffold 3. PP_0844 iscA (Q88PK6) — A-type carrier 4. PP_0845 hscB (Q88PK5) — co-chaperone 5. PP_0846 hscA (Q88PK4) — chaperone 6. PP_0847 fdx (Q88PK3) — ferredoxin

Medium priority (resolve ambiguity/over-annotation): 7. PP_1969 (Q88LG4) & PP_2482 (Q88K11) — confirm/deny MoaA function 8. PP_2435 iscS-II (Q88K56) & CsdA (Q9Z408) — assign specific desulfurase roles 9. PP_1233 tusA-I (Q88NH6) — confirm whether relay TusA or paralog

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8. Evidence base and open questions

Direct (species-specific) evidence: All locus/domain assignments are from PSEPK sequences (proteome UP000000556) + InterPro/Pfam. The contiguous PP_0841–PP_0847 operon structure is direct genomic evidence for ISC completeness, and the absence of `sufABCDSE` is a direct negative genomic result. A published PSEPK study confirms a TudS-type [4Fe-4S] desulfidase recycling 4-thiouridine monophosphate in KT2440 (P 37891428), corroborating active tRNA-sulfur metabolism.

Transferred (homology / related-organism) evidence: The mechanistic ISC assembly model and the TusA→TusBCD→TusE→MnmA relay architecture derive from *E. coli* biochemistry (P 39870763; P 16387657). Transfer to PSEPK is **strong** because the domain compositions and operon structures are conserved, but the specific in-vivo roles of the paralogous desulfurases (PP_2435, CsdA) and MoaA-like proteins (PP_1969, PP_2482) are **uncertain** in PSEPK.

Open questions / experiments to resolve gaps: - Which desulfurase (IscS vs. PP_2435 vs. CsdA) donates sulfur to ISC vs. Moco vs. the tRNA relay in PSEPK? (targeted knockouts + Fe–S/thiolation phenotyping) - Are PP_1969/PP_2482 true Moco enzymes or other radical-SAM enzymes? (complementation of a *moaA* mutant) - Is ISC essential/inducible under oxidative stress given the SUF absence? (conditional depletion; IscR regulon mapping)

Key references

- PMID **39870763** — Gervason et al. 2025. Step-by-step [2Fe-2S] assembly by the *E. coli* ISC machinery (IscU/IscS/Fdx). Defines ISC core.
- PMID **39632806** — Steinhilper et al. 2024. Ferredoxin binding to the core ISC complex (electron donor role of Fdx).
- PMID **16387657** — Ikeuchi et al. 2006. Sulfur-relay for 2-thiouridine: IscS→TusA→TusBCD→TusE→MnmA reconstitution.
- PMID **37891428** — Fuchs et al. 2023. A KT2440 gene product acts as a TudS [4Fe-4S] desulfidase in vivo (direct PSEPK evidence).
- UniProt proteome **UP000000556** (PSEPK); InterPro/Pfam domain signatures (queried Iterations 1–2).

Uncertainty statement: Conclusions about operon completeness and SUF absence are high-confidence (direct genomic). Specific in-vivo roles of paralogous desulfurases, MoaA-like radical-SAM proteins, and DsrEFH/TusBCD are homology-based inferences requiring experimental confirmation in *P. putida* KT2440.
